OPEN PROBLEMS IN NUMERICAL LINEAR ALGEBRA

RA-14

Bounds and a spectral-to-PCA reduction

Exact two-sided matrix–vector queries

Status: partial research result, not a complete solution.
The proofs below give a universal rank lower bound, an exact padding reduction, and matching bounds in a sufficiently-large-dimension regime that allows the rank to grow. They do not characterize the query complexity for every simultaneous choice of dimension, rank, and accuracy.

Main reduction. On a symmetric input with positive leading eigenvalues and a suitable gap, an already valid spectral rank-k approximation can be converted into a strong PCA approximation with at most

$$k\left\lceil \frac{10}{\sqrt{\varepsilon}} \right\rceil$$

additional matrix–vector products. A weighted graph inequality and a Fejér-type polynomial remove a logarithmic accuracy loss from a simpler subspace-angle argument.

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Generated mathematical research note. The reduction proofs are written out; the two cited complexity theorems are imported. No claim of priority, independent peer review, or machine-checked proof is made. The accompanying code is a numerical illustration, not an oracle lower-bound certificate.

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1 Problem, scope, and results

For $n \geq 2$, $1 \leq k < n$, and $0 < \varepsilon < 1/2$, an unknown $A \in \mathbb{R}^{n\times n}$ can be queried through either Ax or $A^T x$. Each vector product costs one query. Queries may be fully adaptive; other computation is free. The required output is $Z \in \mathbb{R}^{n\times k}$ with $Z^T Z = I_k$ and

$$\mathbb{P}\Big[\Big\|A(I - ZZ^T)\Big\|_2 \leq (1+\varepsilon)\sigma_{k+1}(A)\Big] \geq 0.99 \quad \text{for every } A. \tag{1.1}$$

The minimum worst-case query budget is $q_{\mathrm{sp}}(n, k, \varepsilon)$. This is the RA-14 formulation in the supplied repository [1]. Algorithms are understood to be measurable, so the probabilities in the model are defined.

All logarithms below are natural. Universal constants are independent of n, k, ε. A block product with b columns always costs b vector queries.

Theorem 1.1 (Universal bounds and a large-rank regime). *For a universal constant $C_{\mathrm{up}} > 0$,*

$$k \;\leq\; q_{\mathrm{sp}}(n,k,\varepsilon) \;\leq\; \min\left\{n, \left\lceil C_{\mathrm{up}} \frac{k\log n}{\sqrt{\varepsilon}} \right\rceil\right\}. \tag{1.2}$$

Consequently $q_{\mathrm{sp}}(n,k,\varepsilon) = \Theta(n)$ whenever $k \geq n/2$, uniformly in ε. On the promised subclass $\operatorname{rank}(A) \leq k$*, exactly k queries suffice and are necessary.*

The lower bound and promised-subclass statement are proved in Section 2. The n-query upper bound simply learns all columns. The Krylov upper bound is the spectral guarantee of Musco–Musco [2, Algorithm 2 and Theorem 10], applied to A^T; the query accounting is given in Section 3.

Theorem 1.2 (Matching bounds with polynomially large dimension). *There exist universal constants $C_*, D, c_* > 0$ such that*

$$n \geq C_* \left(\frac{k}{\varepsilon}\right)^D \quad \Longrightarrow \quad c_* \frac{k\log n}{\sqrt{\varepsilon}} \;\leq\; q_{\mathrm{sp}}(n,k,\varepsilon) \;\leq\; \left\lceil C_{\mathrm{up}} \frac{k\log n}{\sqrt{\varepsilon}} \right\rceil. \tag{1.3}$$

In particular, this statement is not restricted to constant k.

The lower bound in Theorem 1.2 is a deduction from the published PCA lower bound of Simchowitz–El Alaoui–Recht [3], using the postprocessing theorem proved in Section 5. The exponent D is not optimized or evaluated explicitly here. The dimension restriction is essential to what has actually been proved in this note.

A further, rank-independent lower bound follows by exact padding:

$$q_{\mathrm{sp}}(n,k,\varepsilon) \geq c_{\mathrm{BN}} \frac{\log(n-k+1)}{\sqrt{\varepsilon}} \quad \text{if } n-k+1 \geq C_{\mathrm{BN}}\varepsilon^{-2.01}, \tag{1.4}$$

with universal constants and an enlarged universal threshold if necessary. This transfers the bounded hard instances of Bakshi–Narayanan [4, Theorems 1.1, 5.2, and Section 6]; the transfer itself is proved in Section 7.

What is not established

Equations (1.2), (1.3), and (1.4) do not match in all remaining parameter regimes. In particular, the proof does not justify placing an n-cap around the high-dimensional lower bound and then asserting it for every n, k, ε. The upper bound permits that cap; the lower-bound construction still has a dimension hypothesis. This is the specific missing step, rather than a claim that the requested problem is impossible to solve.

2 A universal lower bound of k queries

The zero-error requirement on rank-k inputs supplies a useful lower bound, but a proof must handle both oracle directions and arbitrary adaptivity. A random orthogonal projector alone is not an adequate construction when k is close to n: one can learn its small complementary subspace instead. We use a nonsymmetric rank-k family.

Lemma 2.1 (Unrevealed Gaussian block)**.** *Let $H \in \mathbb{R}^{n\times k}$ have independent standard Gaussian entries. A deterministic adaptive procedure observes products Hc and $H^T x$. Let $C \subset \mathbb{R}^k$ and $X \subset \mathbb{R}^n$ be the spans of the respective query vectors, with dimensions r and s. Conditional on the transcript,*

$$H = \bar{H} + P_{X^\perp} G P_{C^\perp}, \qquad \bar{H} = P_X H + H P_C - P_X H P_C, \tag{2.1}$$

where $\bar{H}$ is known and the remaining block is a fresh standard Gaussian map from $C^\perp$ to $X^\perp$. Equivalently, G can be taken to have fresh independent standard Gaussian entries in fixed ambient coordinates.

Proof. Initially the statement is the definition of H. Conditional on a transcript, the next query vector is fixed. A new column query depends, in the unrevealed block, only on its component in $C^\perp$. If that component is nonzero, choose it as the first vector of an orthonormal basis of $C^\perp$. The query reveals one column of an independent Gaussian matrix, leaving all other columns independent and unchanged. A row query has the same argument with the first row of the remaining block. A redundant query reveals nothing. These steps prove the assertion by induction, including adaptive choices of the next vector. The two known restrictions are HP_C and $P_X H$; inclusion–exclusion gives the displayed $\bar{H}$. □

Lemma 2.2 (Independence of the row responses)**.** *Before k total nonredundant queries have been made,* $\operatorname{rank}(H^T X) = \dim X = s$ *almost surely.*

Proof. Consider a new row query $x \notin X$. Conditional on the preceding transcript, $H^T x$ has a nondegenerate Gaussian component with full support in $C^\perp$, of dimension $k-r$. Since $r+s<k$, this dimension exceeds s. The affine support of the new answer cannot be contained in the span of the old s row answers. Its probability of lying in that span is therefore zero. The rank increases by one. Column queries do not alter the already obtained row responses. Induction proves the claim. □

Proposition 2.3 (Adaptive two-sided rank lower bound)**.** *No algorithm making fewer than k vector queries can satisfy* (1.1) *on every rank-k input.*

Proof. Set

$$A = \begin{bmatrix} H^T \\ 0_{(n-k)\times n} \end{bmatrix}. \tag{2.2}$$

Almost surely, $\operatorname{rank} A = k$. An A query reveals $H^T x$; an A^T query reveals Hc, where c consists of the first k coordinates of its input. Thus these are exactly the queries in Lemma 2.1. Because $\sigma_{k+1}(A) = 0$, success requires

$$\operatorname{range} Z = \operatorname{range} H.$$

Fix the algorithm's random seed, and condition on its transcript after $q<k$ queries. Write $S = \operatorname{range} Z$, $N = X^\perp$. We have $r+s \le q < k$.

If $N \not\subset S$, choose a nonzero $c \in C^\perp$. By (2.1), Hc has a Gaussian distribution with affine support parallel to N. A proper intersection of that support with S has probability zero. Hence $\mathbb{P}[\operatorname{range} H \subset S \mid \text{transcript}] = 0$.

If $N \subset S$, Lemma 2.2 gives $P_X \operatorname{range} H = X$. Consequently $\operatorname{range} H + N = \mathbb{R}^n$. The inclusions $\operatorname{range} H \subset S$ and $N \subset S$ would imply $S = \mathbb{R}^n$, contradicting $\dim S = k < n$. Success is again impossible.

Thus success has probability zero for this input distribution after fixing any seed. Integrating over the seed gives the same conclusion for randomized algorithms. An algorithm succeeding with probability 0.99 on every input would also succeed with that probability under this distribution, a contradiction. ☐

Corollary 2.4 (Exact complexity under a rank promise)**.** *On inputs promised to satisfy* $\operatorname{rank} A \le k$*, the query complexity is* k*.*

Proof. Take k independent Gaussian vectors g_j and query $A^T g_j$. Their span equals the row space of A almost surely. Orthonormalize and, if the actual rank is less than k, extend to a k-dimensional subspace. The residual is zero. Proposition 2.3 supplies necessity. ☐

3 Upper bounds and vector-query accounting

Querying $Ae_1, \ldots, Ae_n$ determines A. A singular-value decomposition then gives a right singular basis satisfying the target with equality. This proves the exact n-query cap without any spectral assumptions.

For the faster upper bound, the right-sided version of block Krylov uses

$$\operatorname{range}\Big[A^T G, (A^T A)A^T G, \ldots, (A^T A)^t A^T G\Big], \qquad G \in \mathbb{R}^{n\times k} \text{ Gaussian.}$$

The cited spectral theorem yields the 0.99 guarantee for $t = O(\log n/\sqrt{\varepsilon})$, with the exact-column branch used when it is cheaper [2]. A reorthogonalized implementation starts with $A^T G$ and alternates products by A and A^T on each newly produced block. It also retains the images AQ needed to form the small Ritz problem. At depth t this costs at most

$$k(2t+2) \tag{3.1}$$

vector products, including compression. No block is charged as a single vector product. Selecting the top right singular vectors of the already known AQ requires no further oracle access.

The underlying polynomial analysis can be written using $\log(n/\varepsilon)$. In the branch where $k \log n/\sqrt{\varepsilon} < n$, one has $\log(1/\varepsilon) \le 2\log n$ after a universal constant adjustment. Outside that branch, learning all columns gives the desired cap. Thus the displayed global upper bound is genuinely finite-dimensional.

Combining this cap with Proposition 2.3 gives $n/2 \le q_{\mathrm{sp}} \le n$ for $k \ge n/2$, proving the large-rank assertion of Theorem 1.1.

4 What a valid spectral approximation says about angles

Lemma 4.1 (Uniform overlap)**.** *Let* $V \in \mathbb{R}^{n\times k}$ *be a top right singular basis of* A*, and let* $a = \sigma_k(A) > 0$*. If* $Z^T Z = I_k$ *and* $\left\| A(I - ZZ^T) \right\|_2 \le \tau < a$*, then*

$$\sigma_{\min}(V^T Z)^2 \ge 1 - \frac{\tau^2}{a^2}. \tag{4.1}$$

Proof. For the leading left singular basis U and diagonal singular block Σ, one has $V^T = \Sigma^{-1} U^T A$. Therefore $\left\| V^T (I - ZZ^T) \right\|_2 \le \tau / a$. Moreover,

$$V^T Z Z^T V = I_k - V^T (I - ZZ^T) V \succeq \left(1 - \frac{\tau^2}{a^2}\right) I_k.$$

The eigenvalues on the left are the squared singular values of $V^T Z$. □

Example 4.2 (A sharp obstruction to direct PCA inference)**.** Take $n = 2k+1$, $a = 1 + 2\varepsilon$, $b = 1$, $\tau = 1 + \varepsilon$, and

$$A = \operatorname{diag}(aI_k, 0_k, b), \qquad Z = \begin{bmatrix} \alpha I_k \\ \sqrt{1-\alpha^2} I_k \\ 0 \end{bmatrix}, \quad \alpha^2 = 1 - \left(\frac{\tau}{a}\right)^2.$$

Then $Z^T Z = I_k$, $\sigma_{k+1}(A) = b$, and

$$\left\| A(I - ZZ^T) \right\|_2 = \max\{b, a\sqrt{1-\alpha^2}\} = \tau. \tag{4.2}$$

Thus the target is met exactly and the angle bound is sharp. However,

$$\frac{\operatorname{tr}(Z^T A Z)}{\sum_{i=1}^k \lambda_i(A)} = \alpha^2 = \frac{2\varepsilon + 3\varepsilon^2}{1 + 4\varepsilon + 4\varepsilon^2} \longrightarrow 0.$$

A spectral approximation need not itself capture nearly all of the leading PCA objective. A PCA lower bound therefore cannot simply be applied to the original output Z without a further argument.

5 Spectral approximation to PCA in $O(k/\sqrt{\varepsilon})$ extra queries

This section proves the main reduction. The starting block is already a valid spectral approximation; the postprocessor does not have to discover such a block from an arbitrary random initialization.

Theorem 5.1 (Deterministic warm-start postprocessing)**.** *Let $M \in \mathbb{R}^{n\times n}$ be symmetric. Its top k eigenvalues are positive, $\lambda_i \ge a$, and all remaining eigenvalues μ_j satisfy $|\mu_j| \le b$. Suppose $0 < \varepsilon < 1/2$,*

$$a \ge (1 + \tfrac{3}{2}\varepsilon) b, \qquad Z^T Z = I_k, \qquad \left\| M(I - ZZ^T) \right\|_2 \le (1+\varepsilon) b. \tag{5.1}$$

There is a deterministic postprocessor, requiring neither a nor b, which uses at most $k\lceil 10/\sqrt{\varepsilon}\rceil$ further products with M and returns $\widehat{V}^T \widehat{V} = I_k$ with

$$\operatorname{tr}(\widehat{V}^T M \widehat{V}) \ge \left(1 - \frac{\varepsilon}{200}\right) \sum_{i=1}^k \lambda_i. \tag{5.2}$$

The same lower bound holds with M replaced by $|M|$ in the objective.

5.1 A weighted graph inequality

Assume $b > 0$ and set $\tau = (1+\varepsilon)b$. Work in an eigenbasis in which $M = \operatorname{diag}(\Lambda_1, \Lambda_2)$, where $\Lambda_1 = \operatorname{diag}(\lambda_i)$ and $\Lambda_2 = \operatorname{diag}(\mu_j)$. Since $a > \tau$, Lemma 4.1 says that the leading block of Z is invertible. Its range is therefore the graph

$$\operatorname{range} Z = \operatorname{range} \begin{bmatrix} I_k \\ F_0 \end{bmatrix}$$

for a matrix $F_0 \in \mathbb{R}^{(n-k)\times k}$. A vector perpendicular to this graph has the form $(-F_0^T y, y)$. Applying the residual bound to these vectors gives

$$F_0(\Lambda_1^2 - \tau^2 I)F_0^T \preceq \tau^2 I - \Lambda_2^2. \tag{5.3}$$

Define the positive definite diagonal matrices $D_1 = \Lambda_1^2 - \tau^2 I$ and $D_2 = \tau^2 I - \Lambda_2^2$. Equation (5.3) is equivalent to

$$F_0 = D_2^{1/2} K D_1^{-1/2}, \qquad \|K\|_2 \le 1, \qquad \|K\|_F^2 \le k. \tag{5.4}$$

Unlike an unweighted angle bound, this controls the amount of leakage into each tail eigenvalue according to its distance from the residual threshold.

5.2 A polynomial and its elementary bounds

For an integer $m \ge 1$, define the polynomial

$$p_m(u) = \frac{T_m(u) - 1}{u - 1}, \qquad p_m(1) = m^2, \tag{5.5}$$

where $T_m(\cos\theta) = \cos(m\theta)$ is the Chebyshev polynomial. The quotient has degree $m-1$. The trigonometric identity

$$p_m(u) = m + 2\sum_{j=1}^{m-1} (m-j)T_j(u) \tag{5.6}$$

implies that p_m is positive and nondecreasing for $u \ge 1$. For $-1 \le u \le 1$ it also gives

$$0 \le p_m(u) \le \min\left\{ m^2, \frac{2}{1-u} \right\}, \tag{5.7}$$

with the second bound omitted at $u = 1$. Indeed, $p_m(\cos\theta) = \big(\sin(m\theta/2)/\sin(\theta/2)\big)^2$. The first bound follows by expressing the sine quotient as a sum of m unit complex numbers, and the second from $|T_m(u) - 1| \le 2$. For $u > 1$,

$$p_m(u) = \frac{\cosh(m \operatorname{arcosh} u) - 1}{u - 1}. \tag{5.8}$$

5.3 Trace loss after filtering

Apply $p_m(M/b)$ to the starting block. Its range is the graph of

$$F = p_m(\Lambda_2/b) F_0 p_m(\Lambda_1/b)^{-1}. \tag{5.9}$$

An orthonormal basis of this graph is $Y = [I; F](I + F^T F)^{-1/2}$. Put $C = (I + F^T F)^{-1}$. The trace loss satisfies

$$\begin{aligned}\sum_i \lambda_i - \mathrm{tr}(Y^T M Y) &= \mathrm{tr}((\Lambda_1 - aI)F^T F C) + \mathrm{tr}(F^T (aI - \Lambda_2) F C) \\ &\leq \sum_{i,j} (\lambda_i - \mu_j)|F_{ji}|^2. \end{aligned} \tag{5.10}$$

For the first inequality use $0 \preceq F^T F C \preceq F^T F$ and $\Lambda_1 - aI \succeq 0$; for the second term use $0 \prec C \preceq I$ and $F^T(aI - \Lambda_2)F \succeq 0$. No commutation of Λ_1 and $F^T F$ is assumed.

Substituting (5.4) and (5.9) into (5.10), and using $\|K\|_F^2 \leq k$, bounds the loss by k times the largest scalar coefficient

$$\mathcal{B}(\lambda, \mu) = \frac{(\lambda - \mu)(\tau^2 - \mu^2)}{\lambda^2 - \tau^2} \left(\frac{p_m(\mu/b)}{p_m(\lambda/b)} \right)^2, \quad \lambda \geq a, \quad |\mu| \leq b. \tag{5.11}$$

For fixed μ, $(\lambda - \mu)/(\lambda^2 - \tau^2)$ is decreasing for $\lambda > \tau$, because the numerator of its derivative is

$$\mu^2 - \tau^2 - (\lambda - \mu)^2 < 0.$$

The denominator polynomial is nondecreasing and positive. It is therefore safe to replace λ by $a_0 = (1 + \frac{3}{2}\varepsilon)b$ when bounding (5.11) from above.

5.4 An explicit constant bound

Choose

$$m = \left\lceil \frac{10}{\sqrt{\varepsilon}} \right\rceil. \tag{5.12}$$

Scale $b = 1$ for the moment, write $e = \varepsilon$, $s = 1 - \mu \in [0, 2]$, and put $a_0 = 1 + \frac{3}{2}e$, $t_0 = 1 + e$. The numerator in (5.11) is

$$N = p_m(\mu)^2 (\tfrac{3}{2}e + s)(e + s)(2 + e - s).$$

If $s \geq e$, inequality (5.7) gives

$$N \leq \frac{4}{s^2} \cdot \frac{5}{2}s \cdot 2s \cdot \frac{5}{2} = 50.$$

If $s \leq e$, then $m\sqrt{e} \leq 10 + \sqrt{e} < 11$, so

$$N \leq \frac{25}{2} m^4 e^2 < \frac{25}{2} 11^4 < 200000.$$

Thus $N < 200000$ in both cases.

For $0 \leq x \leq 1$, the power series gives $\cosh x \leq 1 + x^2$. It follows that $\mathrm{arcosh}(1 + \frac{3}{2}e) \geq \sqrt{e}$. Also $a_0^2 - t_0^2 = e + \frac{5}{4}e^2 \geq e$. Consequently,

$$p_m(a_0)^2 (a_0^2 - t_0^2) \geq \frac{(\cosh 10 - 1)^2}{(9/4)e} > \frac{4 \cdot 10^8}{9e}. \tag{5.13}$$

The elementary numerical inequality used here has the exact certificate

$$\cosh 10 - 1 > \sum_{j=1}^{7} \frac{10^{2j}}{(2j)!} = \frac{5860808350}{567567} > 10000.$$

Combining numerator and denominator, then restoring b, proves

$$\mathcal{B}(\lambda, \mu) < \frac{9}{2000} eb < \frac{eb}{200}. \tag{5.14}$$

Hence the filtered graph satisfies

$$\sum_i \lambda_i - \mathrm{tr}(Y^T M Y) \le \frac{k\varepsilon b}{200} \le \frac{\varepsilon}{200} \sum_i \lambda_i.$$

5.5 Implementing the proof without spectral side information

The vector block $p_m(M/b)Z$ lies in

$$\mathcal{K}_{m-1}(M, Z) = \mathrm{range}[Z, MZ, \ldots, M^{m-1}Z]. \tag{5.15}$$

The algorithm builds this space and returns its top k algebraic Ritz vectors. The variational principle ensures that their trace is at least $\mathrm{tr}(Y^T MY)$. The algorithm does not evaluate p_m or use the unknown b; the polynomial only proves that a sufficiently good subspace is present.

To count products, orthogonalize successive frontier blocks. Each frontier has at most k columns. Compute its image under M, retain that image for the compression, and use its component outside the current space as the next frontier. Process frontiers numbered 0 through $m-1$, including the last image block needed for the Ritz matrix. At most km products are used. If the space becomes invariant or fills $\mathbb{R}^n$, stop earlier. All needed entries of $Q^T MQ$ are already known.

Finally, $|M| - M \succeq 0$, giving the claim for $|M|$. If $b = 0$, the residual in (5.1) is zero and Z already spans the leading space; the same algorithm can stop after detecting invariance. This proves Theorem 5.1.

6 The growing-rank lower bound in large dimension

6.1 The imported theorem, with its dimension condition retained

We use the following consequence of [3, Theorem 1, Proposition 2.1, and Theorem 2.2]. For $0 < g \le 1/2$, set

$$\lambda = \frac{1 + \sqrt{g(2-g)}}{1-g}, \qquad L = \lambda + \lambda^{-1}, \qquad L - 2 = gL, \qquad M = W + \lambda UU^T, \tag{6.1}$$

where U is uniform in $\mathrm{Stief}(n, k)$ and W is independent GOE, with off-diagonal variance $1/n$ and diagonal variance $2/n$. Define

$$\mathcal{E}_\gamma = \{\|W\|_2 + (1-\gamma)gL \le \lambda_k(M)\} \cap \{\lambda_1(M) \le (1+\gamma)L\}. \tag{6.2}$$

There are universal $c_0, c_2 > 0$ and polynomial dimension thresholds such that, under the law conditioned on $\mathcal{E}_{1/2}$, every fully adaptive algorithm using $T \le c_0 k \log n/\sqrt{g}$ has probability at most $\exp(-n^{c_2})$ of returning an orthonormal $\widehat{V}$ satisfying

$$\mathrm{tr}(\widehat{V}^T |M| \widehat{V}) \ge (1 - g/12) \sum_{i=1}^{k} \sigma_i(M). \tag{6.3}$$

For fixed $\gamma = 1/20$ and failure probability 0.01, a further polynomial threshold gives $\mathbb{P}(\mathcal{E}_{1/20}) \ge 0.99$. These results are imported, not re-proved here.

6.2 Checking every hypothesis of the postprocessor

Set $g = \min\{4\varepsilon, 1/2\}$, so $\varepsilon \le g \le 4\varepsilon$. On $\mathcal{E}_{1/20}$, put $a = \lambda_k(M)$, $b = \sigma_{k+1}(M)$, and $w = \|W\|_2$. Interlacing for the positive rank-k perturbation gives $-w \le \lambda_j(M) \le w$ for all $j > k$. Also $a > w$ on this event, so the leading k eigenvalues are positive and are exactly the leading singular values. Furthermore,

$$a - b \ge \frac{19}{20} gL, \qquad a \le \lambda_1(M) \le \frac{21}{20} L,$$

and consequently

$$\frac{b}{a} \le 1 - \frac{19}{21} g. \tag{6.4}$$

If $b = 0$, the postprocessor's conclusion is immediate. Otherwise, when $\varepsilon \le 1/8$,

$$\frac{a}{b} \ge \frac{1}{1 - 76\varepsilon/21} \ge 1 + \frac{76}{21}\varepsilon > 1 + \frac{3}{2}\varepsilon.$$

When $\varepsilon \ge 1/8$,

$$\frac{a}{b} \ge \frac{42}{23} > \frac{7}{4} \ge 1 + \frac{3}{2}\varepsilon.$$

Thus any successful output of an RA-14 algorithm satisfies all hypotheses of Theorem 5.1. Its postprocessed output meets (6.3), since $\varepsilon/200 < g/12$.

6.3 Probability and query budget

Suppose an RA-14 algorithm uses at most q queries. On symmetric M, either oracle direction is just a product with M, so the same algorithm is admissible for the imported theorem. Append the deterministic postprocessor. The total budget is at most

$$T = q + k \left\lceil \frac{10}{\sqrt{\varepsilon}} \right\rceil. \tag{6.5}$$

Since $\mathcal{E}_{1/20} \subset \mathcal{E}_{1/2}$,

$$\mathbb{P}(\mathcal{E}_{1/20} \mid \mathcal{E}_{1/2}) \ge 0.99.$$

The original approximation algorithm succeeds with probability at least 0.99 for every fixed M, even when we condition on a particular generating pair (W, U). Therefore the composed PCA success probability under the conditioned hard law is at least $0.99^2 = 0.9801$.

For large enough dimension, this contradicts the imported theorem if the right side of (6.5) is at most $c_0 k \log n/\sqrt{g}$. Hence

$$q > c_0 \frac{k \log n}{\sqrt{g}} - k \left\lceil \frac{10}{\sqrt{\varepsilon}} \right\rceil. \tag{6.6}$$

Since $g \le 4\varepsilon$ and $\lceil 10/\sqrt{\varepsilon} \rceil \le 11/\sqrt{\varepsilon}$, a fixed universal condition $\log n \ge 44/c_0$ absorbs the overhead and gives

$$q \ge \frac{c_0}{4} \frac{k \log n}{\sqrt{\varepsilon}}.$$

All imported dimension conditions are polynomial in k and $1/g$. Together with the fixed threshold just used, they are implied by $n \ge C_*(k/\varepsilon)^D$ for universal C_*, D. This completes the proof of Theorem 1.2.

Remark 6.1 (What the reduction does and does not remove)**.** The postprocessor eliminates a possible $\log(1/\varepsilon)$ overhead from a cruder angle-amplification proof. It does not eliminate the dimension hypothesis of the imported PCA construction. Nor does the subtraction in (6.6) by itself prove a useful bound in every small dimension.

7 Exact padding: transferring a hard lower rank

Lemma 7.1 (Rank-padding extraction without loss)**.** *Let $p \geq 0$, let $B \in \mathbb{R}^{m\times m}$, and set $A = \operatorname{diag}(LI_p, B)$. Suppose $Z \in \mathbb{R}^{(p+m)\times(p+r)}$ has orthonormal columns and $\left\|A(I - ZZ^T)\right\|_2 \leq \tau < L$. Then an orthonormal $U \in \mathbb{R}^{m\times r}$ can be computed from Z without queries such that*

$$\left\|B(I - UU^T)\right\|_2 \leq \tau. \tag{7.1}$$

Proof. Write $Z = [C; D]$, where C has p rows. If C did not have full row rank, there would be a nonzero vector $(a, 0)$ perpendicular to $\operatorname{range} Z$. Its norm is multiplied by L under A, contradicting $\tau < L$. Thus $\dim\ker C = r$. Let N have orthonormal columns spanning $\ker C$ and put $U = DN$. Then $U^TU = N^TZ^TZN = I_r$.

For $x \perp \operatorname{range} U$, we have $N^TD^Tx = 0$ and hence $D^Tx \in \operatorname{range} C^T$. Choose a with $C^Ta = -D^Tx$. The vector (a, x) is perpendicular to $\operatorname{range} Z$, so

$$L^2 \|a\|^2 + \|Bx\|^2 \leq \tau^2(\|a\|^2 + \|x\|^2).$$

Since $L > \tau$, this implies $\|Bx\| \leq \tau\|x\|$. Taking the supremum over $x \perp \operatorname{range} U$ proves the result. The case $p = 0$ is the same assertion with $U = Z$. ☐

7.1 Application to the bounded rank-one hard family

Let $m = n - k + 1$, $p = k - 1$, and $r = 1$. On the hard rank-one family used for (1.4), $\|B\|_2 < 2$ and $\sigma_2(B) = 1$. Use $A = \operatorname{diag}(4I_{k-1}, B)$. Each query to A or A^T can be simulated using one query to the corresponding oracle for B; the other block is known. Also $\sigma_{k+1}(A) = \sigma_2(B) = 1$.

On every successful run, $\tau = 1 + \varepsilon < 4$, so Lemma 7.1 extracts a valid rank-one output for B at the same accuracy and with no additional query. On a failure run the reduction may return an arbitrary unit vector if extraction is undefined. Thus the success guarantee is preserved, proving (1.4) from the cited rank-one lower bound. This argument uses a known norm bound on the hard family; it does not assume that an arbitrary unbounded matrix can be scaled for free without access to its norm.

8 Why several tempting extensions are insufficient

Multiplying a rank-one lower bound by k. A direct sum of k visible hard blocks is not a direct-sum theorem for this oracle. A single query has components in all blocks and queries them all in parallel. For example, if each visible block is a rank-one orthogonal projector, one random product reveals a spanning vector of each block's range simultaneously. A proof against all adaptive algorithms needs an additional argument before a factor of k is valid.

Confusing a valid residual with a good PCA objective. Example 4.2 meets the spectral requirement while its captured PCA fraction tends to zero. The weighted postprocessing theorem repairs this specific obstruction; quoting a PCA theorem alone does not.

Confusing subspace depth with vector-query count. A lower bound on the number of block iterations is not automatically a lower bound multiplied by a chosen block size for every adaptive algorithm. Both the upper-bound implementation and the postprocessor above explicitly count individual vector products.

Taking the minimum with n on both sides. An exact-learning algorithm supplies an upper cap. It does not convert a lower bound valid under $n \geq C_*(k/\varepsilon)^D$ into a lower bound outside that region. Doing so would discard a quantified hypothesis of the proof.

The remaining mathematical task

A complete answer to RA-14 needs matching, universally quantified bounds in the regimes not covered by Theorem 1.2 or $k \geq n/2$. For example, the current argument does not resolve arbitrary intermediate powers of k relative to n, nor accuracy regimes that violate the imported dimension threshold. This note supplies no replacement hard distribution or all-adaptive argument that closes those gaps. It also does not claim the known global upper bound is the sharp expression in every such regime.

9 Reproducible numerical checks

The package contains an oracle-counted implementation of the exact-column method, promised-rank recovery, right block Krylov, and the symmetric Ritz postprocessor. Dense residual norms and eigenvalues are computed only by validation code, not inside the oracle algorithms. The mathematical model uses exact real arithmetic; the code uses double precision.

All 18 unit tests passed in the supplied run. They cover query accounting, rank-deficient cases, adaptive Gaussian completion identities, exact symbolic polynomial identities and constants, the sharp counterexample, padding, and the noncommuting weighted trace inequality. A finite set of tests does not verify a universal statement over adaptive algorithms.

9.1 Block Krylov trials, including observed failures

For $n = 96$, $k = 4$, and $\varepsilon = 0.1$, each row below summarizes 30 trials with independent random left and right orthogonal bases. The target residual ratio is 1.1. No theorem-certified universal depth constant is inferred from these trials.

Spectrum family	Depth	Queries	Successes	Worst ratio
Flat spike	2	24	12/30	1.199993
Flat spike	4	40	29/30	1.199745
Flat spike	8	72	30/30	1.000227
Slow decay	2	24	30/30	1.089883
Slow decay	4	40	30/30	1.020983
Slow decay	8	72	30/30	1.000000
Large dynamic range	2	24	30/30	1.001784
Large dynamic range	4	40	30/30	1.000000
Large dynamic range	8	72	30/30	1.000000

The flat-spike family has four singular values equal to 1.2 and a bulk from 1 down to 0.05. Slow decay uses 0.96^j. The third family starts with $10^4, 10^3, 10, 2$ and then decreases from 1 to 0.01. All failures are retained in `results/krylov_trials.csv`.

9.2 Warm starts and scalar checks

Sixty rotated instances were generated from the contraction representation (5.4), which ensures the starting residual bound. The postprocessor met its trace target in all 60 trials; the smallest final trace fraction was 0.9999999999999971. No query budget was exceeded. These small examples often converge much more accurately than the proved bound, and some Krylov spaces fill the ambient space.

Twenty scalar grids, with ε from 10^{-8} to 0.49, tested (5.11). The largest observed coefficient divided by ε was approximately $3.472 \cdot 10^{-10}$, below the proved $1/200$ bound. The grid is only a numerical check; Section 5 supplies the continuous-domain proof. The maximum orthogonality error across the 270 block-Krylov trials was $6.242 \cdot 10^{-15}$.

9.3 Commands and limitations

From the extracted package directory:

```
python -m pip install -r requirements.txt
OPENBLAS_NUM_THREADS=1 python -m unittest discover -s tests -v
OPENBLAS_NUM_THREADS=1 python run_experiments.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The recorded environment was Python 3.13.5, NumPy 2.3.5, and SymPy 1.14.0. Floating-point results can vary slightly with the numerical library. There is no formal verification in a proof assistant. The imported asymptotic constants are not estimated by the experiments.

10 Conclusion

The note proves the global rank lower bound and the $\Theta(n)$ regime $k \geq n/2$, and develops an exact padding transfer. Its main additional argument is a spectral-to-PCA postprocessor costing $k\lceil 10/\sqrt{\varepsilon}\rceil$ products. Combined with a published adaptive PCA lower bound, this gives matching $\Theta(k \log n/\sqrt{\varepsilon})$ bounds under a polynomial large-dimension hypothesis, with growing k allowed.

This does not fully solve RA-14. The unresolved part of this write-up is the simultaneous finite-parameter characterization outside those regimes. The source files, proof dependencies, numerical checks, and this limitation are included together so that the established reductions can be inspected without mistaking them for a complete answer.

References

[1] Open Problems in Numerical Linear Algebra. *RA-14: Optimal query complexity of spectral rank-k approximation.* Repository problem statement, accessed September 12, 2026. https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/randomized-and-low-rank-approximation/RA-14.

[2] Cameron Musco and Christopher Musco. *Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition.* 2015. arXiv:1504.05477. https://arxiv.org/abs/1504.05477.

[3] Max Simchowitz, Ahmed El Alaoui, and Benjamin Recht. *Tight Query Complexity Lower Bounds for PCA via Finite Sample Deformed Wigner Law.* STOC 2018; arXiv:1804.01221v2, 2020. https://arxiv.org/abs/1804.01221.

[4] Ainesh Bakshi and Shyam Narayanan. *Krylov Methods are (nearly) Optimal for Low-Rank Approximation.* 2023. arXiv:2304.03191v1. https://arxiv.org/abs/2304.03191.